



Assignment 3 (35 pts)

Due April 22, 2025

1. (2pts) Consider the following optimization problem with a single equality constraint

$$\begin{aligned} \text{Minimize:} \quad & x_1^3 + x_2^3 \\ & x_1 \in \mathbb{R}, x_2 \in \mathbb{R} \\ \text{Subject to:} \quad & x_1 + x_2 = 1 \end{aligned}$$

- (1pts) What are the first-order KKT conditions for this problem?
 - (1pts) Find a point that satisfies the first-order KKT conditions and verify that it is indeed a local minima.
2. (4pts) Consider the following quadratic programming problem with two equality constraints

$$\begin{aligned} \text{Minimize:} \quad & 3x_1^2 + 2.5x_2^2 + 2x_3^2 + 2x_1x_2 + x_1x_3 + 2x_2x_3 - x_1 - x_2 - x_3 \\ \text{Subject to:} \quad & x_1 + x_3 = 3 \\ & x_2 + x_3 = 0 \end{aligned}$$

- (a) (1pt) Write the first-order optimality conditions for this optimization problem. [Hint: You can rewrite the objective function in matrix quadratic form to simplify the derivation of optimality conditions.]
 - (b) (3pts) Numerically calculate the optimal solution by solving the first-order optimality equations (you may need to write a small MATLAB script for calculating the numerical solution).
3. (3pts) Consider the constrained optimization problem

$$\begin{aligned} \text{Minimize:} \quad & x_1(1 - x_2^2) \\ & x_1 \in \mathbb{R}, x_2 \in \mathbb{R} \\ \text{Subject to:} \quad & x_1^2 + x_2^2 - 1 = 0 \end{aligned}$$

- (a) (1pt) Find the optimum solution using the first-order KKT conditions.
- (b) (2pts) Now consider the case when the equality constraint is rewritten as

$$(x_1^2 + x_2^2 - 1)^2 = 0.$$

Can you find the optimum solution of this modified problem using the first-order KKT conditions? Explain your steps and reasoning clearly.

4. **(3pts)** Consider the optimization problem arising in robust design applications

$$\min_{\mathbf{x} \in \mathbb{R}^n} J(\mathbf{x}),$$

where $J : \mathbb{R}^n \rightarrow \mathbb{R}$ denotes Bayes risk which is defined as

$$J(\mathbf{x}) = \int f(\mathbf{x}, \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta},$$

where $\boldsymbol{\theta}$ denotes a vector of d random variables with joint probability density function $p(\boldsymbol{\theta})$, and $f : \mathbb{R}^n \times \mathbb{R}^d \rightarrow \mathbb{R}$ is a black-box function.¹

- (a) **(1pt)** What computational challenges do you anticipate when solving this class of optimization problems?
- (b) **(2pts)** Which of the optimization methods we covered in AER1415 would you consider applying to this class of problems? Justify your choice of numerical method.
5. **(10pts)** Consider the NASA airfoil noise dataset $\mathcal{D} := \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^N$, where $\mathbf{x} \in \mathbb{R}^5$ contains the input settings used during wind tunnel experiments and $y \in \mathbb{R}$ denotes the scaled sound level measurement.² Our goal is to fit a simple neural network model to approximate y as a function of $\mathbf{x}$, i.e., $\hat{y}(\mathbf{x}; \boldsymbol{\theta}) = \mathbf{W}_2 \tanh(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + b_2$, where $\mathbf{W}_1 \in \mathbb{R}^{m \times 5}$, $\mathbf{b}_1 \in \mathbb{R}^m$, $\mathbf{W}_2 \in \mathbb{R}^{1 \times m}$, and $b_2 \in \mathbb{R}$ are the model parameters. We compactly denote the model parameters by $\boldsymbol{\theta} \in \mathbb{R}^n$, where $n = 7m + 1$).

The optimization problem for estimating the model parameters using the N experimental observations in $\mathcal{D}$ can be formulated as

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^n} \frac{1}{N} \sum_{i=1}^N (y^{(i)} - \hat{y}(\mathbf{x}^{(i)}; \boldsymbol{\theta}))^2.$$

A training dataset with $N = 1000$ observations for estimating the model parameters is provided in `airfoil_training_data.dat`. You are provided a MATLAB/Python function in `loss.{m,py}` that returns the loss function value and its gradients with respect to the model parameters given a batch of observations. To illustrate the steps involved in estimating the model parameters, you are provided a baseline parameter estimation workflow that uses the conjugate gradient method in `airfoil_train_script.{m,py}`. You are also provided a testing dataset containing additional experimental measurements in `airfoil_testing_data.dat` (distinct from the training dataset) for estimating the generalization error.³

¹The functional form of f is unknown and therefore the integral cannot be computed analytically.

²T.F. Brooks, D.S. Pope, and A.M. Marcolini. Airfoil self-noise and prediction. Technical report, NASA RP-1218, July 1989. <https://ntrs.nasa.gov/api/citations/19890016302/downloads/19890016302.pdf>.

³See `airfoil_train_script.{m,py}` for details on how to use the testing dataset and the prediction function provided in `model_predict.{m,py}` to evaluate model performance. The script also provides an initialization strategy for the model parameters which you can use in your numerical studies.

- (a) **(3pts)** Implement the stochastic gradient descent (SGD) algorithm to estimate the parameters of the neural network model. Use the function provided in `loss.{m,py}` to calculate the loss function value and its gradients. Your SGD implementation should be able to handle mini-batches of data and you can assume that the learning rate is constant.
 - (b) **(5pts)** Estimate the model parameters by minimizing the least-squares loss on the training data provided in `airfoil_training_data.dat` with 1000 points. Use a neural network with $m = 64$ for all experiments and run for 1000 epochs without employing any additional termination criterion. Experiment with different batch sizes (e.g., $\{16, 64\}$) and learning rates (e.g., $\{0.001, 0.01, 0.1\}$), and report the statistics of the mean square error (MSE) values on the training and testing datasets.⁴ Discuss your findings and the effect of the batch size and learning rate on the convergence of the loss function and the model performance. Provide representative convergence plots and tables to summarize your results and findings.
 - (c) **(2pts)** Compare the performance statistics of the models fitted using your SGD implementation to models obtained using the baseline conjugate gradient (CG) method provided in `airfoil_train_script.{m,py}`. Use at least 10 runs for different random number seeds to evaluate performance. Discuss the differences in convergence behavior and generalization error of models fitted using SGD and CG.
6. **(7pts)** Design an optimization algorithm that hybridizes the PSO algorithm with the MATLAB constrained optimization solver `fmincon`.⁵ The naïve approach for hybridization would be to run the PSO algorithm for several iterations to generate a good initial guess for `fmincon`. You are expected to formulate a more sophisticated hybrid search strategy that synergistically combines both algorithms. You can either formulate an original algorithm or implement a hybrid algorithm published in the literature.
- (a) **(3pts)** Describe the details of your algorithm (along with a compact pseudocode), including user-defined parameters (if any) in the report.
 - (b) **(4pts)** Implement the hybrid optimization algorithm you are proposing in MATLAB/Python and apply your code to the 20-dimensional constrained bump test function (**P3**). You can reuse the code from Assignment 1 for this. Compare the performance statistics of the hybrid algorithm to the standard PSO algorithm developed in Assignment 1.

⁴Perform at least 10 independent runs with different random seeds (see the initialization strategy for the model parameters in `airfoil_train_script.{m,py}`) for at least 6 different settings of batch size and learning rate. For making predictions using your model, use the function provided in `model_predict.{m,py}`. Report mean $\pm$ standard deviation of MSE error over the training and testing datasets.

⁵If you are using Python for your assignments, please use an appropriate algorithm in `scipy.optimize` such as `SLSQP`.

7. (6pts) In manufacturing processes, determining appropriate tolerances for design variables is crucial. Tight tolerances increase production costs, while loose tolerances risk performance degradation or constraint violations. This problem addresses the fundamental engineering trade-off of establishing maximum permissible deviations from optimal design specifications that still guarantee acceptable performance bounds. Such tolerance analysis is essential in industrial design practice, where balance between production feasibility and functional requirements must be rigorously established.

Consider an engineering design problem which involves the solution of the following constrained minimization problem:

$$\mathbf{x}^* = \arg \min_{\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^n} f(\mathbf{x}),$$

where $\mathcal{X} := \{\mathbf{x} : g_j(\mathbf{x}) \leq 0, j = 1, 2, \dots, m\}$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ denotes the system performance metric (objective function), and $g_j : \mathbb{R}^n \rightarrow \mathbb{R}, j = 1, 2, \dots, m$ denote a set of inequality constraints.

You are tasked with estimating quality control targets for $\mathbf{x}^*$ (i.e., the *maximum* permissible deviations for $\mathbf{x}^*$) such that

- (a) the worst-case value of the performance metric is $(1+\eta)f(\mathbf{x}^*)$ with $\eta = 0.05$ and
- (b) the inequality constraints are always satisfied.

Formulate an optimization problem whose solution will yield the maximum permissible deviation in $\mathbf{x}^*$ (quality control targets). Clearly define all optimization variables and any new variables that you introduce in your formulation.

[Hints and suggestions: This is an open-ended problem with multiple possible solutions. As a first step, you will need to construct a sensible parametrization for the quality control target. For example, the quality control targets could be modeled as an interval of the form $[\mathbf{x}^* - \boldsymbol{\delta}, \mathbf{x}^* + \boldsymbol{\delta}]$, where $\boldsymbol{\delta} \in \mathbb{R}^n$. To begin, consider the simpler case when there are no constraints before tackling the case with inequality constraints. You may also consider an approach that simultaneously solves for $\mathbf{x}^*$ and the quality control targets.]

Bonus (5pts) Implement the SGD algorithm with momentum to estimate the parameters of the regression problem in Q5. Compare the performance statistics of the SGD algorithm with momentum to the standard SGD algorithm and the baseline method based on CG in `airfoil_train_script.m.py`. Discuss your findings and the effect of momentum on the convergence of the algorithm. Provide representative plots and tables to summarize your results.

Submission guidelines: Submit an **electronic copy** of your report in pdf format along with **documented** MATLAB/Python scripts and **input files** (if any). You should include a file named “README” outlining how the scripts should be run. Upload a single tar or zip file containing all files to Quercus.⁶

⁶Please verify the integrity of your tar/zip file before uploading it to Quercus. Do not use proprietary compression formats such as RAR.